

QMS Journal Club

Choong Pak Shen

February 8, 2026

QMS Journal Club 4th meeting: Pure and mixed states

- ① A brief review on linear algebra
 - Basic operations on vectors and matrices
 - Pauli matrices
 - Inner product and outer product
 - Tensor product and Kronecker product
 - Trace of a matrix
- ② Postulates of quantum mechanics
 - State space of a closed quantum system
 - Dynamics of a closed quantum system
 - Quantum measurement
 - Composite quantum system
 - Superdense coding

Consider a vertically polarized photon, given by

$$|V\rangle = \frac{|L\rangle + |R\rangle}{\sqrt{2}}. \quad (1)$$

When the photon passes through a circular polarizer that allows either only $|L\rangle$ or only $|R\rangle$ polarized light, half of the photons will be absorbed in both cases.

This may seem like half of the photons are in $|L\rangle$ or $|R\rangle$, but this is not true.

If the photon passes through a linear polarizer, there will be no absorption.

Meanwhile, if we pass unpolarized light through any type of polarizer, half of the intensity of light will be lost.

For the case of polarized light, the action of “measuring” the polarization of light is contextual, i.e. the outcome depends on what kind of measurement is being performed.

For the case of unpolarized light, the action of “measuring” the polarization of light is not contextual, because the outcome does not depend on the type of measurement being performed.

This suggests that for the case of polarized light, a quantum treatment is required (contextuality is a quantum phenomenon), whereas a classical treatment is required for the case of unpolarized light.

Polarized light is one example of pure states, whereby the probability of obtaining $|L\rangle$ or $|R\rangle$ is given by the absolute square of the probability amplitude.

Unpolarized light is one example of mixed states, and it is considered as a (classical) statistical ensemble, i.e. each photon having either $|R\rangle$ or $|L\rangle$ polarization with probability $\frac{1}{2}$.

For this example,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (2)$$

These two ensembles are completely indistinguishable experimentally, hence they are considered the same mixed state.

Conventionally,

$$|H\rangle = |0\rangle, |V\rangle = |1\rangle, \quad (3)$$

$$|D\rangle = |+\rangle, |A\rangle = |-\rangle, \quad (4)$$

$$|L\rangle = |+i\rangle, |R\rangle = |-i\rangle. \quad (5)$$

Formally, the density matrix of a quantum system is defined as

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|, \quad (6)$$

where $\sum_i p_i = 1$. We say that ρ is a mixture of pure states.

Density matrix is a more general formalism compared to state vector formalism, since a pure state $|\psi\rangle$ can be described by a density matrix $\rho = |\psi\rangle \langle \psi|$.

Consider

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \quad (7)$$

$$|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle, \quad (8)$$

$$\rho = p_\psi |\psi\rangle \langle \psi| + p_\phi |\phi\rangle \langle \phi|, \quad (9)$$

where $p_\psi + p_\phi = 1$. Find ρ in a 2×2 matrix.

The density matrix is given by

$$\rho = \begin{pmatrix} p_\psi |\psi_0|^2 + p_\phi |\phi_0|^2 & p_\psi \psi_0 \bar{\psi}_1 + p_\phi \phi_0 \bar{\phi}_1 \\ p_\psi \bar{\psi}_0 \psi_1 + p_\phi \bar{\phi}_0 \phi_1 & p_\psi |\psi_1|^2 + p_\phi |\phi_1|^2 \end{pmatrix}. \quad (10)$$

If we compute $\langle 0|\rho|0\rangle$, we get $p_\psi |\psi_0|^2 + p_\phi |\phi_0|^2$.

The diagonal element gives the probability of finding a particle in the corresponding basis state $\{|0\rangle, |1\rangle\}$.

For a general state $|\Psi\rangle$,

$$\begin{aligned} \langle \Psi|\rho|\Psi\rangle &= p_\psi \langle \Psi|\psi\rangle \langle \psi|\Psi\rangle + p_\phi \langle \Psi|\phi\rangle \langle \phi|\Psi\rangle \\ &= p_\psi |\langle \psi|\Psi\rangle|^2 + p_\phi |\langle \phi|\Psi\rangle|^2 \end{aligned} \quad (11)$$

it can be seen that $\langle \Psi|\rho|\Psi\rangle$ gives the total probability of finding a particle in the pure state $|\Psi\rangle$ within a mixture ρ .

A density matrix ρ has the following properties:

- 1 $Tr(\rho) = 1$;
- 2 $\rho^\dagger = \rho$;
- 3 It is a positive semi-definite matrix, i.e. $\langle \psi | \rho | \psi \rangle \geq 0$.

For pure states, the density matrix ρ further satisfies the following:

- 1 $\rho^2 = \rho$;
- 2 $Tr(\rho^2) = 1$.

How many real parameters do you need to describe a 2×2 density matrix?

You need three real parameters to describe a 2×2 density matrix.

Consider again the one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle. \quad (12)$$

Its density matrix is given by

$$\rho = \begin{pmatrix} \cos^2\left(\frac{\theta}{2}\right) & e^{-i\phi} \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi} \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) & \sin^2\left(\frac{\theta}{2}\right) \end{pmatrix} \quad (13)$$

Previously, we learn that

$$P_x = \langle\psi|X|\psi\rangle = \sin\theta \cos\phi, \quad (14)$$

$$P_y = \langle\psi|Y|\psi\rangle = \sin\theta \sin\phi, \quad (15)$$

$$P_z = \langle\psi|Z|\psi\rangle = \cos\theta. \quad (16)$$

We want to rewrite the density matrix in terms of $\{P_x, P_y, P_z\}$, which are called the polarization (Bloch) vectors,

$$\begin{aligned}
 \rho &= \begin{pmatrix} \frac{\cos \theta + 1}{2} & \frac{\sin \theta (\cos \phi - i \sin \phi)}{2} \\ \frac{\sin \theta (\cos \phi + i \sin \phi)}{2} & \frac{1 - \cos \theta}{2} \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} P_z + 1 & P_x - iP_y \\ P_x + iP_y & 1 - P_z \end{pmatrix} \\
 &= \frac{1}{2} (I + P_x X + P_y Y + P_z Z) \\
 &= \frac{1}{2} (I + \vec{P} \cdot \vec{\sigma}), \tag{17}
 \end{aligned}$$

where

$$\vec{P} = \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix}, \quad \vec{\sigma} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \tag{18}$$

In general, a density matrix can be written as $\rho = \frac{1}{2} (I + \vec{r} \cdot \vec{\sigma})$.

Density matrix formalism is useful when

- 1 the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
- 2 when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).